Fall Monitoring System for Elderly Care: Project Proposal

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I. Problem Statement and User Need

As people age, day-to-day tasks that were once routine become increasingly difficult, yet many older adults choose to live independently despite this. An elderly person living alone who falls has no reliable way of summoning help, and caregivers or family members have no way of knowing something happened until it's too late.

This gap affects three primary groups directly: the elderly individual living alone, who bears the immediate risk of a fall, the caregiver or family member directly responsible for their well-being, and the care facility that needs to keep constant track of its patients. Table I lists the needs of every stakeholder:

TABLE I
Provisional stakeholder needs.

ID	Need
User (Wearer)	
N1	"If I fall and can't get up, I need help to come without me having to do anything."
N2	"I want to be able to cancel an alert if I am fine before anyone is bothered."
N3	"I don't want to feel as though I am constantly being watched."
Caregiver / Family	
N4	"I need to be notified immediately if something happens."
N5	"I want to know if the wearer is incapacitated after a fall."
Care Facility	
N6	"I need alerts to be routed reliably so that response protocols are followed."
Personal	
N7	"The device firmware needs to be reproducible on new devices without editing the source."
N8	"I don't want too much money to be spent on developing this device."

Taken together, these needs point to one measurable target for the system as a whole, stated below as the project objective:

Objective – The fall detection system should be able to detect at least 95% of real falls (miss rate $\leq 5\%$) and deliver a confirmed alert to the caregiver/family and care facility within 25 seconds.

II. Scope and System Context

Meeting these needs requires deciding upfront what this system will and will not attempt this term. Table II lists what is built against the nearest capability deliberately left out, so that the boundary of the prototype is explicit.

TABLE II
Scope of the proposed prototype.

In Scope	Out of Scope
Wrist IMU-based fall detection (watch).	Detecting other health events (seizures, cardiac events, etc).
Waist IMU and barometric sensing (custom board).	Additional sensors (impact, heart rate, microphone, etc).
Low-fidelity prototype development (breadboard/protoboard).	Custom PCB fabrication, enclosure, volume manufacturing costing and economics.

Figure 1 shows the system as a single boundary containing the fall-detection system, with no detailed internal structure drawn at this stage. The wearer's motion crosses into the system, and a confirmed alert crosses out to a computer running an MQTT broker, which pushes the notification to a caregiver-facing dashboard. Two further stakeholders, the caregiver/family member and the care facility, sit downstream of the dashboard and are each reached independently.

Two stakeholders behind this diagram want different things from the same design decision. The wearer's success criterion is retaining the ability to cancel a false trigger before anyone is bothered (N2); the caregiver's is being notified the instant a fall occurs, with no delay standing between a real emergency and their response (N4). These two criteria collide directly: any nonzero cancel window granted to the wearer is time subtracted from how quickly the caregiver can be alerted to a genuine fall. No single window length fully satisfies both, and finding a middle-ground is one of the provisional requirements addressed in Section IV.

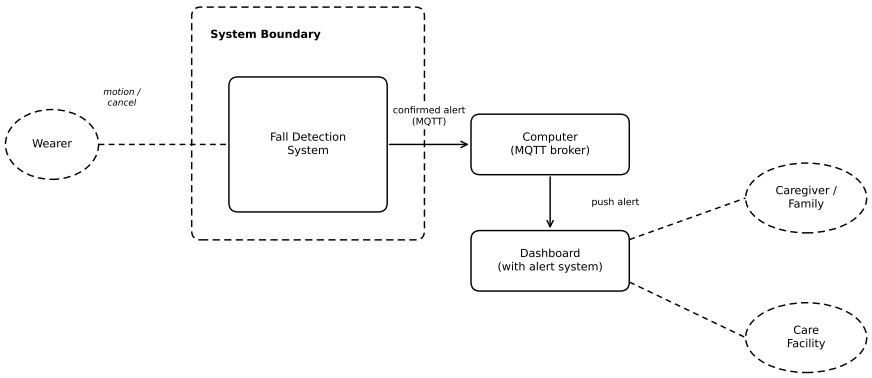


Fig. 1. System context diagram. The wearer's motion enters the system for processing, and is then moved downstream to the remaining stakeholders.

III. Platform and Custom Hardware

The watch imposes several fixed limits on this design. Table III lists only the specifications this project actually touches, each tagged with its source and marked as a constraint.

TABLE III
Platform limits touched by this project, tagged as constraints.

ID	Limit	Source	Rationale
C1	6-axis IMU (wrist)	Vendor wiki	Fixed sensing hardware the wrist-side fall classifier runs on.
C2	Bluetooth 5 (LE)	Vendor wiki	Fixed radio the watch to PCB validation link and alert-to-broker path depend on.
C3	2.4 GHz Wi-Fi (802.11 b/g/n)	Datasheet	Fixed radio for the MQTT alert path to the computer.
C4	ESP32-S3R8, dual core, up to 240 MHz	Vendor wiki	Fixed processing budget the on-device fall classifier must run within, in real time.

The custom hardware is a waist-worn board carrying its own microcontroller with an integrated BLE radio, a 6-axis IMU, and a barometric pressure sensor powered independently of the watch. Since it is worn separately rather than physically attached, it cannot reach the watch's I2C, UART, or USB expansion pads; instead, the two devices exchange fall-event and validation data wirelessly over Bluetooth Low Energy. This link is budgeted within REQ-FD-02 in Section IV, which bounds how quickly a candidate fall event must reach the watch for validation as part of the overall delivery time.

IV. Provisional Requirements

Table IV lists eight provisional requirements derived from the needs in Section I, covering all five requirement classes. Each is marked provisional, since none has yet been tested; Section V treats the numeric bounds still resting on unvalidated assumptions as uncertainties to be resolved before Stage Gate 1.

Trial counts, thresholds, and disclaimers. Rate and timing-based criteria (REQ-FD-02, REQ-FD-03, REQ-FD-04) use 20 trials per category, since at this count each trial corresponds to a 5% increment, matching the resolution of the stated miss-rate threshold; binary pass/fail requirements (REQ-FD-01, REQ-FD-05, REQ-FD-06) use 10 repetitions, enough to catch an intermittent failure without requiring the statistical power a rate claim needs. Every numeric bound above is derived from a stated assumption or an existing figure in the table rather than asserted without reasoning, but none has yet been validated by measurement. REQ-FD-07's $200 cap applies **only to this term's PoC-stage proof and prototyping**. Continued development in Project II incurs an added budget cost, not estimated here as it is not within the scope of this semester's work.

V. Uncertainties and the PoC Plan

Three uncertainties are ranked below by how much of the design would collapse if the underlying assumption proved false. Each is written as a register entry; the single PoC plan attacks the highest-ranked uncertainty first.

PoC Plan. The highest-ranked uncertainty, R1, is attacked first, since it would most undermine the design if wrong. *Question:* On a wearable device, does combining multiple independent acceleration, angular rotation, and pressure signals reliably distinguish real falls from ordinary movement, and does it deliver a notification in time, even over a poor connection? *Measure:* (a) false-alarm and miss rates on held-out data from a labeled set of scripted falls and confusable activities (sitting heavily, lying down, stairs), held out by session and, where possible, by person; (b) detection-to-dashboard

TABLE IV
PROVISIONAL REQUIREMENTS.

ID	Class	Statement	Rationale	Method/Acceptance	Need
REQ-FD-01	Func.	An alert shall be initiated on fall, no wearer action required.	A fall can incapacitate the wearer instantly, help can't depend on their own input (N1).	Demo: alert on simulated fall, 10 trials.	N1
REQ-FD-02	Perf.	Alert shall be delivered within 25 s, including over a degraded link.	25 s = 15 s cancel window (REQ-05) + 10 s delivery margin; board/watch sub-link budgeted to 200 ms (<1% of total).	Test: detection-to-receipt Δ, 20 trials, strong and weak link; board/watch Δ separately.	N1,4,6
REQ-FD-03	Perf.	Miss rate shall be $\leq 5\%$ across tested activities.	Missed falls are life-threatening compared to false alarms; 5% matches 95% target.	Test: false negatives $\div$ fall trials, 20 trials on held-out set.	N1,4
REQ-FD-04	Perf.	False-alarm rate shall be $< 10\%$ across tested activities.	Loose enough to avoid suppressing true detections (REQ-03); still bounds nuisance interruptions to roughly 1 in 10 activities.	Test: false positive $\div$ non-fall trials, 20 trials on held-out set.	N3
REQ-FD-05	Interf.	User shall be able to cancel a pending alert via a button within 15 s.	Fixed independently as the wearer-reaction window, 15 s of the 25 s budget (REQ-02). Resolves wearer/caregiver collision (Sec. II).	Test: trigger-to-press Δ, 10 trials; cancels within window.	N2
REQ-FD-06	Func.	Severity shall be escalated after 60 s of no post-fall motion.	Roughly $\sim 3\times$ a given 10–20 s stumble-recovery time.	Test: Δ to escalation, 10 trials; fires/withholds correctly.	N5
REQ-FD-07	Constr.	Added PoC/prototyping hardware cost shall be $\leq$ \$200.	Rough BOM estimate for PoC/prototype costs (MCU+BLE, IMU, barometer, battery, passives). Scoped to current project.	Inspect: receipts $\leq$ PoC/prototype budget cap.	N8
REQ-FD-08	Build.	Firmware shall build/flash from a clean checkout, no edits.	Reproducibility, commonly forgotten class. No ambiguity between developers.	Demo: build succeeds on 2nd machine, no intervening.	N7

delivery time and success rate at strong and weak link strength, and the system's behavior when the link is lost. *Pass criterion:* the miss rate is at most 5% (REQ-FD-03), the false-alarm rate is below 10% (REQ-FD-04), and every confirmed fall is delivered within 25 s (REQ-FD-02) in all link conditions tested.

TABLE V
RANKED UNCERTAINTY REGISTER.

ID	Statement	Assumption	Consequence	Prob.	Evidence Required	Mitigation
R1	May not meet miss/false-alarm targets on new people/sessions.	Thresholds tuned on development sessions carry over.	Falls go undetected, or the wearer stops wearing the device.	Likely	Rates on held-out sessions with one or more new people vs. pass criterion.	Retune combination rule or widen training set.
R2	60 s of inactivity may not separate incapacitation from deliberate rest.	No motion for 60 s is a meaningful incapacitation proxy.	REQ-06 falsely escalates or fails to escalate.	Likely	Scripted "rest" vs. "recover" trials, check escalation.	Retune window based on measurements during testing.
R3	BLE latency on a moving body may exceed the 200 ms sub-budget (REQ-02).	Bench latency is representative of worn, moving use.	Total delivery exceeds 25 s, or time split needs revision.	Unlikely	Timestamp Δ, both devices worn, in motion.	Re-split budget, or tune BLE connection parameters.

VI. ASSUMPTIONS, DEPENDENCIES, AND THE LOGBOOK

Assumptions. This design rests on several claims taken as true but not yet tested:

- Falls performed by healthy volunteers under scripted, controlled conditions are assumed to produce IMU and barometric signatures representative of an actual elderly fall.
- Bluetooth Low Energy connectivity between a wrist-worn watch and a waist-worn custom board is assumed to remain reliable at typical in-home separation distances.
- The watch's processing budget (C4) is assumed sufficient to run the on-device fall classifier and validation logic within the real-time constraints implied by REQ-FD-02's 200 ms board-to-watch sub-budget.

Dependencies. This design also counts on conditions outside the project's own control:

- The caregiver, family member, and care facility each have a working internet connection capable of receiving an MQTT-forwarded alert.
- The wearer consistently wears and charges both the watch and the custom PCB; the system cannot function if either device is left uncharged or unworn.

AI-Assisted Steps. An AI assistant was used throughout drafting to help interrogate needs into measurable requirements, classify requirements by type, check scope and needs for internal consistency, and review wording against the course guide's rules.

Key numbers in this proposal trace to logbook entries L-01 through L-11.